

ActInf Textbook Group ~ Cohort 1 ~ Meeting 21

(Chapter 9, part 1)

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hey everyone it is October 21st 2022 we're in cohort one the textbook group it's meeting 21 and we're having our first discussion on chapter nine chapter nine is model based data analysis let's look at what the section headers are there's a short introduction a discussion of meta Bayesian methods which is going to be very interesting and in some ways is even like an entry point to thinking about entity modeling in active inference and so that'll be kind of fun we'll return to variational Laplace resonating with our just completed discussion on chapter four and Laplace then section 9.4 9.5 are going to help us see where data in terms of like gigabytes of actual good data Jacob actual data from measurements and so on where these come into play with models so how does one go from a unfurnished trainable model to a parameterized specified model which actually is explaining variance in real world data sets and then there's some examples of GMS and some models of false inference um well I added only one General question from um a lighter reading I think as we all hear move through this we can generate a lot of other key points and questions um ultimately the models described in this book are only useful if they can answer scientific questions but just we could rehearse the introduction again but let's just go Um into it so metabasian methods this chapter deals with utility of active inference formulations and analyzing data from behavioral experiments so one could imagine all kinds of bodily verbal digital Behavior this goes beyond proof of principle simulations we've seen in previous chapters and instead exploits active inference in answering broadly speaking there's two related reasons for fitting a computational model to observe Behavior the first is to estimate parameters of Interest the second is to compare alternative hypotheses so to parameterize from data is one opportunity that is to make some model of brain function and understand within one person or one group or two groups how you can consider those parameterizations to be phenotypes phenotype is something of a biological organism or system that's measurable so like femur length is a phenotype but also a phenotype doesn't have to just be something that's measurable on the body with a ruler like phenotype might be distance ran

in the first 50 seconds after the hawk flies through the sky on a cloudy day and so then that is still a measurement that could be inferred or discussed or made and we're talking about computational phenotyping because the data and observations for sure are like a basal phenotype like the button being clicked was measurable pheno means to show but also we could talk about like the Precision variables in our cognitive model as being phenotypic and also model comparison can be used so whereas this first modality of parameterization is like the data set is fixed we collected 240 fmri data sets now we're going to parameterize so we're going to give some um plasticity to our model and we're going to fit parameters so that so that our model is like the best resembling it can to these 240 fmri data sets we have in the second modality we're treating in some ways the models as fixed and then evaluating to what extent different models Stack Up and that can be used at a very fine scale to look for different models that fit better to a given data set but also this is where we can talk about like actual biological explanations and predictions so like to give an example from yes please Ali first uh sorry a related question to this whole discussion is uh I think it was in section 9.2 uh it says this goes beyond the proof of principle simulations we've seen in previous chapters and instead exploits active inference and answering scientific questions uh I didn't quite understand this statement here because uh in the previous chapters uh we were we were also engaged with modeling uh the real scientific phenomena right so what does proof of principle simulation here exactly mean and how does it uh differ from the approach uh I mean from answering a real scientific questions yeah good question does anyone want to give a thought or I can give a thought too I mean literally just these following that the first and second reasons to use active inference in this way it's like instead of just modeling something and saying oh that looks like what what we intended the model or oh gosh it looks really predictive or whatever getting more rigorous with it to specifically you know enumerate parameters for a particular model right and and then compare multiple models like trying to say which one is you know uh more precise that's that's a different thing than what has was being done in the book previously like we're just building up the model and seeing how it kind of worked before and now we're like talking about putting it into a like a Machinery that's going to you know uh select on models like which ones best fit or whatever well said I I totally agree um preview although arguably like the chapter five neurobiology the tables of examples were answering scientific questions but those generative models were not largely presented but in the papers one could have gone in but yes the previous generative models were like built from the ground up and then like oh look at this interesting Behavior now we're building the Machinery to take in real data and fit

that and then there's the the two uses of parameterization or the two uses of data fitting to find the best parameters for a given data set and model and also to compare models model comparison all right now we get to the very excellent figure 9.1 so oh is the meta Bayesian inference what does anyone see in this figure Ali well as pointed out in the text uh the inner box somehow represents the generative model of the phenomena we're trying to we're observing but the outer dashed box is somehow our generative model as The Observers so the term metabasian I think used here to denote this kind of change of perspective to higher uh to a higher level of viewpoint uh I mean to add another layer of hierarchy to the already hierarchical generative models we had before yeah great well said and this out so we we could have had an arbitrarily nested um well just to make a simple point this is the discrete time formulation pomdp figure 4.3 and that can be nested in an arbitrary way and composed and meta Bayesian is like putting the wrapping paper on the gift because this outermost layer is us Brock so the last time this came up was uh in the context I think it was in the other cohort but we were talking about the you know provenance of the model the the impetus for it being you know as a tool for uh diagnosis of patients for you know non-typical sort of neuro um so uh I'm I guess I'm curious maybe uh if there's a simple way to understand it from that perspective but I also have um a another question I guess which is If This Is Us modeling it seems like it would be like infinite really recursive because every time you add another layer like then there's another layer that's not being modeled of how we're actually modeling it and therefore we would have to but then if we account for that then then there would be another outer uh you know I don't know if that's just uh infinite regression towards the hidden state one or the other or but um or I'm just that's nonsense but yeah no no it's it's not I think let's let's talk about the uh setting of the clinical neuroimaging the inner model is going to be the cognitive behavioral model of the patient just speaking of this the participant the recipient of action as the patient they're selecting green or blue coins or whatever they're doing some behavioral Paradigm isocating or choice or decision making we're modeling their action perception attention and so on whether we formalize it or not the experimenter's behavior is influencing the parameterization of the patients like we are choosing populations and sampling patients here is like the parameterization of the experiment we'll have 20 people they'll come in for two sessions we're gonna do 11 um you know of this and here's like the data coming out of so this o here is the observation that the patient sees and their inference on what they're seeing experimental stimuli are actually an action from the experimenter we're pushing the o to their screen so this o tilde is their o sequence we're getting data from the experiment we could

choose to be ignorant or implicit about our parameterization or maybe even this is a nested model we made a governance decision in the lab to decide who gets to decide this and the constraints were the cost and the availability nested modeling of the Outer Loop and what about infinite recursion and I think one silver lining or Saving Grace or whatever or just strength of the Bayesian graph or active inference framework is we know we can always Dive In and like build from a node or we can just treat the edges of our base graph as like a Markov blanket with the unknown so we could say we made a simple model of decision making here and then we could this could be a multi-agent simulation and each of those could have another nested model and then we could get it fixed then maybe it's irrelevant or maybe we want to know anyways but what good will it I don't know yeah or it doesn't fit and we do want to know or yeah let's just say that the pragmatic value for us is um High statistical power and sensitivity for a diagnosis of a um neuro condition so like we want the inner loop to be in the point we don't want to be like spending money Beyond diminishing returns on our experiment we want to actually like diagnose and help the people in this limited situation then one could do model comparison with just I'm ignoring all of this I'm just going to parameterize just just I'm going to call it like I see it then one could have another but but if one were to engage in a research program where we we collected one data set then yeah let's just call like we see it but now we have the opportunity to design a second cohort of fmri should we do two people for a hundred sessions or a hundred people for two sessions who should those hundred people be so at that point having a policy selection outer loop enabling seems pretty relevant now we could make that um simple maybe it's we always pick 30 participants maybe it's we either do a small experiment with three or we do a big one with 30. it's just simpler those are the two Pathways we have or maybe it's a you know Etc but then you could do model comparison and evaluation on those increasingly complex outer loop models none of that would be influencing the structure of the cognitive model of the patient for all you block friends heads in the chat one could imagine that there's human fmri cognitive model this is a versionable cognitive model open source and then different lab groups could have open or closed models with open or closed data around their decision making about how they're using a core model this is also just a on a more um qualitative point and then Ali like this is where we can start to talk about moving beyond implicit biases not just like perch implicit biases or anything like that it's just like we're doing investigation and exploration we're externalizing and exteriorizing our priors and we're working such that the relevant features that are in the Box are all externalized in terms of the scientific apparatuses parameterization and that's at least something where we can

move towards and someone could say well you haven't made your hyper hyper hyper priors explicit but isn't that a better conversation to have than you didn't say why you chose this many people for the experiment it could be like you didn't State The Meta governance of how you came to the decision to do 30 I mean it makes it it sounds like it makes it really a lot easier to think about comparing and Counting showing that you accounted for your methods um like it's the same you're doing the exact same experiment right um might be able to move the needle a little bit on some of the reproducible Earth replication crisis things like in that vein yeah great point Ollie I don't see what uh Rock just mentions um as a side note uh I think approaching this um kind of I mean this approach to uh model based or uh better the marble-based data analysis takes the infinitism stance as an implicit uh somehow the philosophical stance because uh if we don't think in terms of infinitism I don't think we were able to model adequately these kind of metabasian reasoning because we'll get at some point to as Brock said just infinite regressive reasons so at least in my opinion it's somehow implicitly takes the infinitism stance here as uh proposed by Peter Klein at least in large part because as infinitism uh stands claims will need somehow the subjectively and objectively available reasons to uh to model or to construct or Justify true beliefs in other words the reason must be uh non-repeating and infinitely available to us in order for us to be able to construct that true belief so I'm not quite sure about it but it looks very similar to infinitism stance yeah interesting um introduction wasn't familiar with this area um like approximate Bayesian computation bounded rationality the composability of Base graphs these are we get to eat our slice of cake and recognize that there is the rest of the cake and the table and the world outside of the restaurant and so on like we can just say the the GM that we made was the thermometer and a temperature value and someone can say but what about humidity and it's like there's a composability but we're not within the positivist or the falsificationist like well my model of temperature and thermometer is positive evidence that that's all that's happening or it's the best model we have we're just waiting for it to be disproven we can have like a Bayesian portfolio of models and all of them can be understood as um Maps as composable Maps could this lead to Bayes optimal experimental design of research programs like statistical power the the famous statistics quotation is like something about an autopsy like the statistician is not going to diagnose it it's they come in and they tell you what went wrong with the autopsy but that's also that's not fundamental to statistics that's actually just a little bit of like a joke about the practice of research which is like the biostatistician just being from my own research experience they're called in for assistance on a complex already collected data set and so it's

like well we had three mice in March and then we had six mice in April how do we balance this versus The Upfront conversation to help design a high statistical power experiment and so like statistics up front it's gonna increase the reproducibility and alignability of experiments it is a sanity check if not a formal verification that the experiment is going to tell you something and it allows for um replicating experiments selecting experiments based upon like their statistical power okay following sections unpack an example of generic inference that may be used for metabasing inference the variational LaPlace with hierarchical models then there's going to be a simple recipe that's related to the uh one of the other questions up here variational LaPlace may be used for more generic likelihood functions than those encountered earlier which were defined as gaussian so previously the family of functions that we are doing variational inference with was gaussians now we can generalize Beyond using only gaussian distributions by saying whatever it is we're going to do a laplacian approximation so even if it's like a something very strange and as mentioned in um cohort 2 discussion just an hour ago the LaPlace approximation has two parameters the the mode the center of the bell and just the width of the parabola it's guaranteed to capture some of the variants of the distribution for distributions with a central tendency gaussian or otherwise it can do well for distributions that are like multimodal it does not do well because it gets tricked to finding the mode which may only capture like a minority of the overall bulk and again the SPM textbook lays out side by side LaPlace variational Bays non-parametric sampling approaches from those laplacian approximations we're going to do variational inference the real distribution might be unfactorizable or not we don't know what family of functions it even is but we know that the LaPlace approximations that we make on those distributions for sure will be amenable to variational inference and because of the quadratic nature we can do gradient Ascent so just like the ball going to the bottom of the bowl this is just an inverted Parabola so it's just gradient asense on the mountain parametric empirical base this usage of a matrix to represent the experimental layout is heavily used in SPM this is looking a lot like a regression this is actually a little bit of a short section 9.4 [Music] this part here comparing the evidence for a model where the second element is allowed to deviate from zero or the precise belief it's zero that is actually very much again like regression testing like if you're gonna if you have um if you're testing for for the effect of a given um Factor on height that is related to the p-value you get for height is related to the model that includes height and the model that doesn't include height and then their sum of squares is compared within the appropriate statistical test family t-test z-test f-score all giving you a p-value so this is very much like statistics on regressions

one thing maybe there there's a deeper reading where this is clear but one way to talk about parametric empirical Bays is that like well first off it is it's parametric you're dealing with parameters so sometimes that's not too helpful but the empirical part means like the way we're going to get our D Matrix are prior is from the data so like let's just say that we have um we're measuring height in the classroom we might be able to use chapter 6 to specify and furnish the model but then as soon as we make that first and then we might set our prior would say well let's have a super loose prior let's say that it could be like uniform across all you know but of course who knows what the maximum height is maybe it's not a human classroom so we can't just specify a uniform distribution across you know every finite value and so a way to kind of break that um a priori challenge where like you want your um prior to be loose enough to accommodate the full range of the possible data but also sharp enough so that you're not just like starting like on like it's just an absurdly uninformative prior all priors are informative even a uniform prior is still informative the thing that matters is the strength the weakness of the prior and the family of the prior so what you could do instead would be you could measure 10 Heights or even one and then use the mean and the variance of that sample empirically to set your prior so set just the mean of the prior on height as the mean that you got from 10 and then you could just over disperse the variance so if there's somebody who's outside of that bound now it's like you know you can be more confident but this is so this is widely used to take an empirical data set and then use the data sets empirical values and summary statistics to parameterize the generative model and then the ball goes from there um and also the expectation maximization algorithm where you have a given data set and you update the parameters in the generative model and then you generate data compatible with that model and continue that is um very close to PEB 9.5 instructions for model based analysis so let's copy these out because the question was like how is a figure 9.2 I guess section 9.5 and it's summarized in 9.2 here's the six steps oh someone added a great Okay so this is a little bit of a review of the textbook but we're starting with data collection maybe the data already have been collected so we're putting aside the situation where we're doing the meta modeling on ourselves and then doing experimental design and then collecting the data we're just going to take it from the data are collected a pomdp again discrete time in this case is structurally prepared according to the recipe in chapter six pomdps would we say are equivalent to specifying a likelihood function or they they embody or they entail a likelihood function what is the arrow between two and three does merely constructing a pomdp in this format uniquely identify a likelihood function or what what work has to happen right here just a thought we

can you know learn yeah uh I don't think it uh these arrows necessarily mean uh that this particular component of uh the modeling process necessarily uh translates perfectly to the other stage uh I think uh they're more likely to be to show the dependencies among each steps so uh in the case of the third stage or the likelihood fun the construction of the likelihood function obviously it depends on what our pomdp model uh the functions so I'm not sure if we can I mean if it will I mean if they were 100 isomorphic I guess they would be redundant right so yeah I think those arrows show dependencies rather than translating into each other thanks I I think I broadly agree and there might also be some other ways to draw it like the prior beliefs is this Theta I think this is using the uh figure nine one ontology this data is Big Theta these are the experimenters parameters the model is the cognitive model of the entity *U tilde* is the observed data so here's the observed data getting passed with pomdp and Associated likelihood function we can the likelihood okay the pmdp describes variables in their relationships the likelihood function allows us to create p which is the distribution of observed Behavior *U tilde* conditioned upon experimental parameters data observations which are in this case the ones that are provided to stimuli experimental stimuli and the model then the parametric empirical Bays comes in when we actually get the data it comes in in six but we're getting there we're on the path when we have the data flowing from the experiment we can then do the model inversion rather than describing the distribution of behavior that would be generated by Theta θ we're going to invert it so we can talk about the distribution of parameters experimental parameters conditioned upon θ Bayesian equation allows us to have a proportionality between what we really want to know this Top Line and I don't know if I can call it a joint distribution but these two multiplied distributions which is the probability of the param of the experimental parameters given the model and separating out the probability of *U H* condition on all the rest so this is leveraging the sparsity of the inverted model to facilitate a form that is amenable to linear regression type equations this is like y equals MX plus b it's not but it is this is like the same structure $y = m \cdot x + b$ but then these bars and their Associated variances which come from a LaPlace this is the p-value and the effect size so for this third modality the effect size is zero and the variance is you know such and such and then for this one we could say the p-value or the base Factor for this Factor mattering it's it's high the evidence for this mattering is extremely high so this is like If This Were a bar chart and we're um if we were in frequency statistics land and it were like 10 plus or minus 1. then we we have a z-score of 10. we're 10 standard deviations away from zero so the p-value of the effect size being greater than zero is whatever it is for a p-value of a z-score

of 10. very low but again this is like where the real data collection is going to be happening and interfacing so now here's their description collect behavioral data formulate the pomdp it takes parameters as input outputs are fully specified but not yet solved pomdp specify likelihood likelihood function specify prior beliefs often these will be centered on zero with precisions reflecting plausible ranges that's interesting note solve for posterior and model evidence Newton is probably a reference there to some gradient derivative base gradient method I'm not exactly sure group level analysis treating the estimated parameters for each individual as if they were generated by a second level model this is exactly structurally Anova the group level analysis of variance we had people throw the ball there was type 1 and type 2 a person and then the the weather type to is there an effective type you make two models one of them is a model without type one of them is a model with two categories of type then those two models can be compared in terms in the frequentest world in terms of their hierarchical likelihood in the Bayesian World however a limitation of the hierarchical likelihood ratio test is that the models must be strictly structurally nested they have to reflect direct simplifications or elaborations of each other in contrast Bayesian modeling allows us to use the Bayes Factor which can compare models that don't need to be strictly nested so you could test very structurally different models against each other it's kind of like information as a common currency in the Bayesian approach it I I considered uh variable one two and three and then someone else could compare variable four five six and you can be like well how good are those two different models against what we care about in terms of informational Criterion AIC Bic the base Factor whereas if somebody said well I did model uh with one two three and the p-value is 0.01 and I did this with four five six and the p-value was 0.01 those can't be directly compared because the p-value is against like a local null hypothesis and there's just like probably other issues so it's one huge advantage of the Bayesian like we're kind of we're past we're 50 years past the Rubicon or 400 or whatever on base but like Bayesian formulations allow for the generative recognition model the tale of two densities so you can specify something that generates and recognizes data that's one advantage another Advantage is that you have access to all the statistical tools of frequentive statistics and you can also access this like informational statistics I'm sure there's like way way way more rabbit holes there but like broadly speaking instead of taking experimental observations in frequentism and then mapping them onto a t distribution only and then saying well now because of the let me publish this paper in the Bayes worlds there are like richer comparisons and like more nuanced penalization functions of how many parameters versus how much variance explains you know

just for example if anyone thought bayesians were too confident or if it wasn't the um leading modern strategy for dealing with uncertainty this design Matrix for a linear model the SPM design matrices are like very interesting looking so the textbook has many many many of these matrices they're encoded on a grayscale so like What's Happening Here here are images through time each row is an image could be an image in a Time series could be like different static images here third column is time so this is time point zero one two three four five six times and then here's condition one and two first condition whether one is white and black it doesn't matter um condition let's just say Y is one condition one was in effect condition two is not and vice versa but you could have overlapping conditions so it's like a graphical representation and like this this is also common to see this kind of like waterfall here's um patient one two three four five Here's the the the four condition the three conditions in a randomized order for the one two three four five patients here's their um Global blood flow here's some other measurements Etc but it turns out with so this summarizes the total design of the experiment and it turns out like you just smash it like a matrix multiplication like the design Matrix multiplied by the observations and one can imagine like if the observations have no relationship with the design Matrix there's a null hypothesis that you could get from random Matrix Theory if the experimental design strongly influences the observations there's some informational outcome 96 in our final few minutes examples of generative models two experiments outlined in nine five the details are not important oh but it's too um isocating models this is a continuous time left tracking a moving Target was a categorical eye tracking as we've mentioned in this list like doing some kind of a webcam eye tracking cognitive model would be massive it's also very you know sensitive technology I think because it would be able to determine somebody's cognitive model and their attention in different ways really be interesting well it's going to say they they actually already do that in specifically for uh like Championship like first person shooter games to try to catch people uh using aimbots like so they try to predict you know where your eyes are going to be based on the data in the game and if you are shooting you know or you're you're acquiring the target faster than your eyes uh you know the quiet could have acquired the target then wow well yeah interesting maybe even some of the um behaviors already are out there so it's not like this is yeah like maybe the data already it's like that moving moving Target one yeah yeah and like interpreting this as like an occupancy density but then having a cognitive model where that is like an uncertainty reduction or a salience or relevance observation which is implicitly what these usages are Disney trailers models of false inference Julius Caesar we had the Obama now we have the

Caesar discussion of phase optimality and disorders different pathological and neurodiverse conditions addiction impulsivity compulsivity delusions hallucinations interpersonal personality disorders ocular motor syndromes pharmacotherapy prefrontal syndromes visual neglect disorders of interoception inference if there's a parameters set that works well there's going to be various parameter sets that exhibit different outcomes and then ask per Foucault Society is the definer of Madness and all of that and the DSM and so you know asterisk asterisk but broadly speaking these are models of pathology we outlined an approach that uses theoretical models described in previous chapters to pose questions to empirical data this lets us use active inference as a non-invasive tool to probe the computational processes that individuals use to make decisions so funny well the model is not invasive I don't think any model is invasive ultimately the six steps in figure 9.1 but it's not that's an error it's figure 9.2 six steps in figure 9.2 provide a generic method for Designing experiments to non-invasively interrogate implicit generative models people or other systems used to drive Behavior that's an opportunity to answer questions about the function of the nervous system in health and indices thank you fellows looking forward to the conversation next week as well thank you peace bye